The Complete Effects of Coors Field: Part 1

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Abstract

Coors Field, home of the Colorado Rockies, has long been recognized as Major League Baseball's most hitter-friendly park, but the full extent of its influence on player performance remains incompletely quantified. In this study, we examine how Denver's extreme altitude affects both pitching and hitting outcomes by comparing performance at Coors Field with all other parks (AOP). Using Statcast's pitch-level and batted-ball data from 2015–2025, we find that the reduced air density at Coors Field's elevation leads to a corresponding 18% decrease in pitch movement, resulting in more desirable pitches for hitters, higher contact rates, and altered launch-angle distributions. Although batters tend to hit at lower launch angles, batted balls in the air travel significantly farther due to the reduced air density. That, with Coors' uniquely large outfield dimensions, leads to an increased rate of hits, especially extra-base hits. Finally, we quantify how these interacting factors collectively increase offensive performance at Coors Field and set the stage for future work on the additional Coors hangover effect and player profile evaluations. This study and those that follow will allow MLB management to make data-driven decisions with player transactions and optimize player usage and overall team strategy in high-altitude parks.

1. Introduction

The goal of this project is to thoroughly analyze the impact Coors Field has on player performance as an exercise of data analysis. This report is meant to demonstrate how we would communicate the key takeaways of our research. If you would like to see the code written to build this report, as well as additional visuals and a more comprehensive account of our thought process, go to our notebook in GitHub.

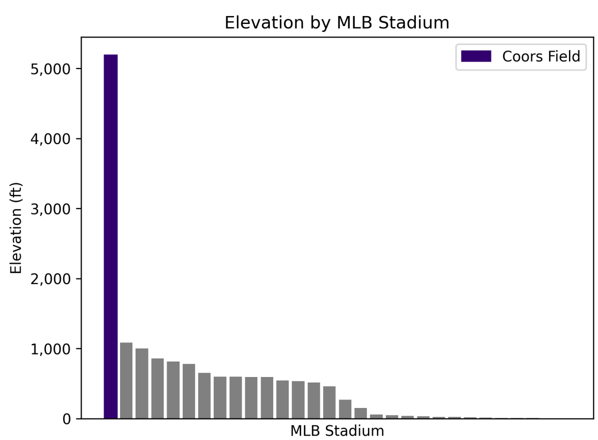

Figure 1. The elevation of all MLB stadiums used in the 2016 season. Other stadiums significantly used from 2015-2025 include Truist Park (elevation 1,001 ft), Glove Life Field (545 ft), Salhen Field (521 ft), Sutter Health Park (24 ft), George M. Steinbrenner Field (34 ft)

2. Coors Effect

Coors Field is located in Denver, Colorado where the altitude is roughly 5,200 feet above sea level, significantly higher than the altitudes of the rest of the MLB parks. This results in the air density at Coors Field being around 18% less than it is at sea level, significantly impacting the way the ball moves, both out of the pitcher's hand and off the bat.

2.1. Out of the Pitcher's Hand

Let's start with how Coors effects movement out of the pitcher's hand. Since pitches move differently depending on the pitcher and pitch type, we grouped pitches by these variables, as well as year to account for changes over time, and split them by when they were thrown at Coors Field vs. all other parks (AOP). Looking at pitchers with at least 100 pitches at both Coors and AOP, we calculated the straight average movement at AOP and how much less the average movement at Coors is.

		Horizontal		Vertical	
Pitch Type	**Pitcher Count**	**Movement (in)**	**Coors Difference**	**Movement (in)**	**Coors Difference**
FF (four-seam fastball)	157	+0.63	-22%	+1.31	-18%
SL (slider)	66	-0.32	-6%	+0.23	-20%
SI (sinker)	60	+1.23	-22%	+0.65	-14%
CH (changeup)	37	+1.05	-21%	+0.60	-19%
FC (cutter)	34	-0.18	-14%	+0.59	-19%
CU (curveball)	24	-0.88	-17%	-0.61	-16%
KC (knuckle curve)	18	-0.46	-32%	-0.66	-24%
ST (sweeper)	9	-1.22	-13%	+0.23	-67%
FS (split-finger)	3	+1.04	-26%	+0.25	-3%

Table 1. Induced Movement by Pitch Type

From this, we can see that pitches move less both horizontally and vertically at Coors. Pitches move due to air resistance (drag) and the Magnus effect, which uses air density to create the force of lift and break. Because of the altitude, air density in Denver is roughly 18% lower than the air density at sea level, though other factors like temperature and humidity have an effect as well. Pitch movement should be directly proportional to air density, and our findings support that.

It's unclear how much Coors Field affects pitch placement. Pitchers almost certainly know about the reduced pitch movement and account for it. Plus pitchers likely account for the heightened offensive production at Coors. Add in the variation in where pitchers aim their pitches and their execution of intended placement, and there's too much noise to make many conclusions.

At Coors Field, swings result in a ball in play roughly 2% more of the time, with about a 3% decrease in both whiffs and foul balls. This may be due to the reduced pitch movement, making it easier to make contact. There was also a slightly increased rate of pitches located in the heart of the strike zone at Coors Field, which would also make it slightly easier to make contact, though this increased rate of pitches down the middle could also be attributed to the reduced pitch movement.

2.2. Off the Bat

Continuing from the effect of Coors on the baseball out of the pitcher's hand and towards the plate, we have the impact off the bat, starting with the final observed effect of the reduced pitch movement.

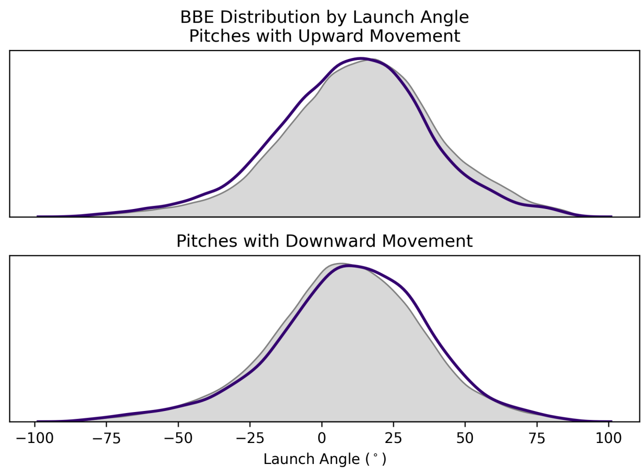

Figure 2. The distribution of launch angles from batted balls.

For pitches with upwards induced movement, the reduced movement at Coors causes the pitches to arrive lower than they appear to be going. As a result, batters tend to get on top of the ball more at Coors, resulting in a lower distribution of launch angles from batted balls at Coors. On the other hand, pitches with downwards induced movement see the opposite effect. The reduced movement at Coors causes these pitches to arrive higher than they appear to be going, resulting in batter getting under the ball and thus a higher distribution of launch angles. Given that the majority of pitches have upwards induced movement, this phenomenon results in more balls in play with low launch angles (i.e. groundballs).

Hitters may tend to get on top of the ball more often at Coors, but when they do put it in the air, the lower air density works in their favor. With less air to slow the ball down, it ends up carrying more.

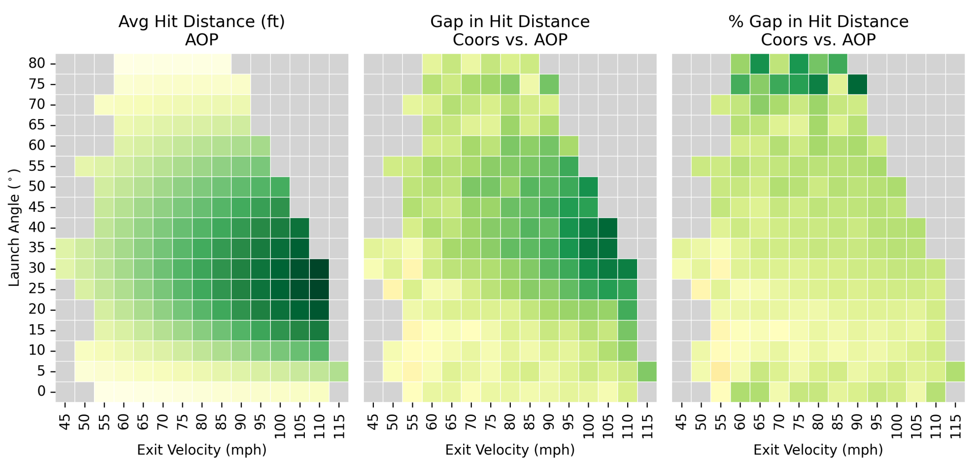

Figure 3. The average hit distance, the difference (in ft) between the hit distance at Coors vs. AOP, and the % gap in hit distance for batted balls with a launch angle of at least 0. Minimum 10 batted balls.

Controlling for the batted ball's launch angle and exit velocity, balls in Colorado consistently travel farther in the air, with a larger gap for balls with further hit distances.

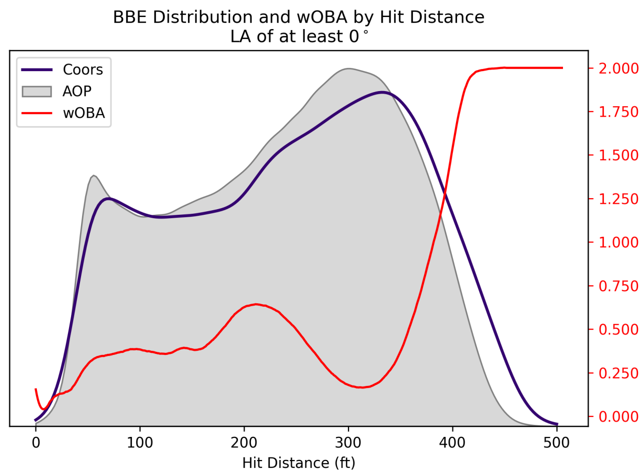

Figure 4. The distribution in the hit distances of batted balls at Coors vs. AOP overlayed with the average wOBA in AOP by hit distance.

So even though batted balls at Coors tend to have lower launch angles, hits at farther distances (around 350 feet and onward) are more common. These hits are valuable as the wOBA line in Figure 4 indicates.

2.3. Park Dimensions

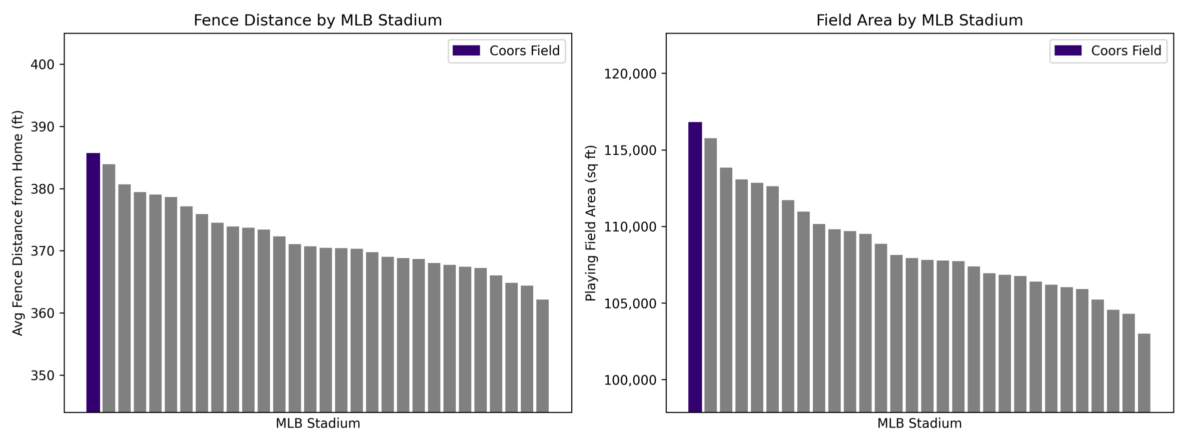

Figure 5. The elevations, average fence distance, and field area of all mlb parks compared to Coors Field.

There may not appear to be any more ways for Colorado's altitude to affect the baseball beyond from pitch to bat and from bat to ground, and indeed there aren't any, directly at least. However, as a way to combat the extra carry from flyballs at Coors, the field dimensions are a bit unique. The outfield walls are pushed back, making the Coors outfield the largest in all of MLB. As a result, the rate of home runs are reduced slightly, though still more than the average stadium. But this has also given outfielders at Coors significantly more ground to cover. While outfielders tend to position themselves around 315 feet away from home plate in AOP, at Coors they're around 330 feet away to compensate for the larger outfield. As a result, batted balls that go around 200-300 feet drop for hits at Coors more often, as the outfielders are farther away. Additionally, while batted balls with a distance around 390-420 feet go for home runs less at Coors than the typical park, they more often go for hits, especially extra-base hits like doubles and triples.

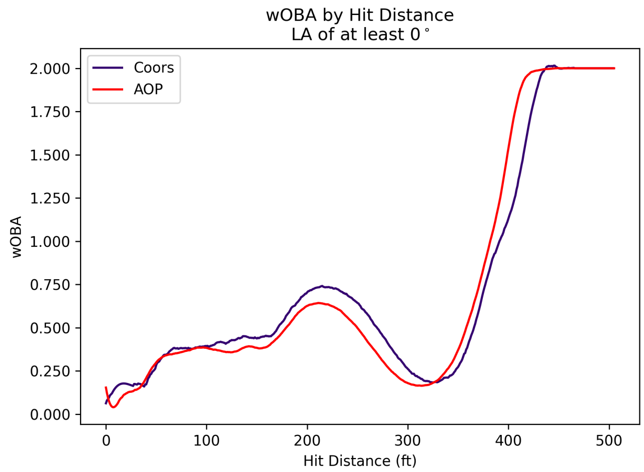

Figure 6. The average wOBA in Coors vs. AOP by hit distance.

The extra distance isn't the only factor at play here, however, as Figure 6 indicates.

2.4. Total Impact

	Coors	**AOP**
BABIP	.325	.290
HR%	3.4%	3.1%
ISO	.238	.209
K%	18.7%	22.7%
BB%	8.1%	7.7%
R/PA	0.200	0.162

Table 2. Offensive performance at Coors vs. AOP for batters on the away team, 2015-2025.

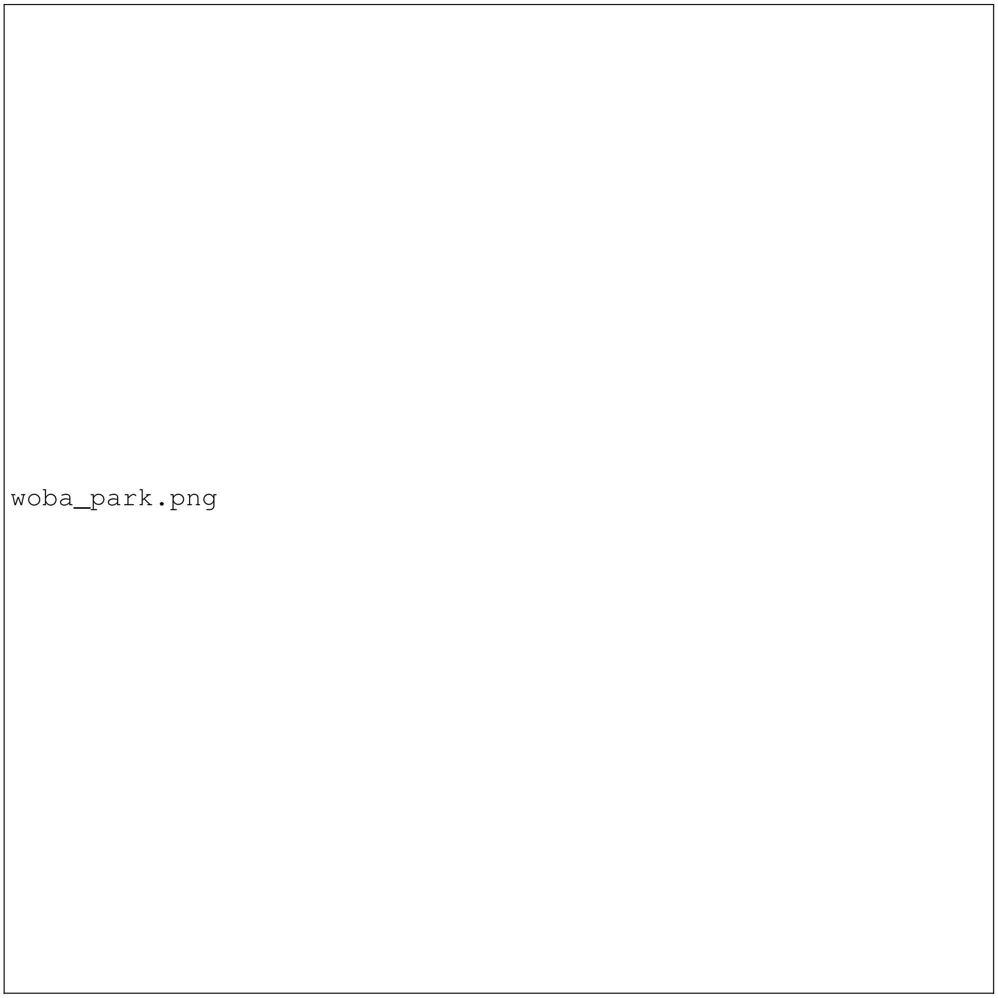

Figure 7. The gap in expected and actual wOBA by home team.

3. Next Steps

References